GCLK Pi Application and Web Portal Requirements

Introduction

The goal of this project is to develop a Python-based application (or an equivalent framework) to operate on a Raspberry Pi CM4 module with the official Raspberry Pi IO Board ("App HW"). The hardware will interface with a custom-designed motor control PCB via USB (serial communication). Additionally, the App HW will communicate over the internet with a custom-developed web application ("Portal") that enables the end client to manage their resources and generate operational reports.

The App HW and motor control board will reside within a custom-built metal kiosk. The primary function of the kiosk is to dispense environmentally friendly laundry sheets, offering students a sustainable alternative to traditional liquid detergent. This system allows students to do their laundry with reduced environmental impact while providing valuable data collection and management capabilities for the client.

Kiosk Overview

The GCLK machine is a custom-built kiosk equipped with a 10.1-inch LCD touchscreen for user interaction. The key components of the kiosk are as follows:

Key Features

1. **Inventory Trays (Cartridges)**:
    - Spring-loaded trays that hold scented and unscented laundry sheets.
    - Present the top sheet to the rollers for dispensing.
2. **Door Switch**:
    - A limit switch that reports whether the door is open or closed.
3. **Motors**:
    - Operate rollers that grab and dispense the sheets through the chute.
4. **Laundry Sheet Chute**:
    - Containers located on the door that catch dispensed sheets, allowing users to retrieve them easily.
5. **Optical Sensors**:
    - Beam-break sensors placed inside the chutes to count dispensed sheets.
    - Sensors are positioned on both the upper and lower sections of the chute for accurate counting.
6. **Time-of-Flight (ToF) Sensors**:

- Two sensors (left and right) measure the distance from the tray bottom to determine inventory levels.
- The measured distance is converted into a percentage for precise inventory tracking.

7. **Limit Switches**:
   - Six switches (three per side) act as backup inventory trackers, complementing the ToF sensors.
8. **10.1-Inch LCD Touchscreen**:
   - Connected to the Raspberry Pi CM4 and serves as the primary user interface for dispensing sheets, conducting surveys, and displaying environmental messages.
9. **Raspberry Pi CM4**:
   - Hosts the application and communicates with both the motor control board and the web application (Portal).
10. **Motor Control Board**:
    - Interfaces with all sensors and controls the motors for dispensing sheets.
    - Communicates with the Raspberry Pi CM4 via USB (serial communication).

Kiosk Application – Python App

The kiosk application provides the graphical user interface and interaction logic for users. There are two distinct versions of the application:

1. Standard Free Application

This version does not require user credentials for dispensing laundry sheets. It is designed to collect usage data and answers to survey questions.

App Flow

1. **Attractor Screen**:
   - Displays a looping GIF when idle.
   - Instructs users to touch the screen to begin.
2. **Dispense Screen**:
   - Prompts users to select scented or unscented laundry sheets.
3. **Survey Question**:
   - Asks users a question (mandatory or optional) before dispensing sheets.
   - The question's requirement is configurable through the Portal.

4. **Environmental Message**:
    - Displays a sustainability message during sheet dispensing for a configurable duration (set in the Portal).
5. **Return to Attractor Screen**:
    - Awaits the next customer interaction.

---

2. Authenticated Application

This version includes an additional step for user authentication, requiring users to enter a **User ID** (email address, phone number, or student ID) after the attractor screen. The authentication method is configurable through the Portal.

App Flow

Attractor Screen → Authentication → Dispense → Survey Question → Environmental Message

---

Web Application – Portal

The web-based Portal serves as a management console for administrators to configure kiosks, generate reports, and monitor their status.

Portal Features

1. **Dashboard**:
    - Displays key status information for the network and resources in a user-friendly format.
    - Provides clickable summary boxes for quick navigation.
2. **Kiosk Management**:
    - Lists all configured kiosks along with:
        - Name
        - Customer
        - Contact details
        - Inventory levels
        - Errors (if any)
        - Status (online/offline)
    - Includes actions for viewing detailed information or editing kiosk settings.
3. **School/Customer Management**:

- Displays a list of customers (educational institutions or other clients).
- Allows administrators to drill down into kiosks owned by specific customers.

4. **Client Management**:
    - Provides a list of users (e.g., students) who use the kiosks to dispense laundry sheets.
5. **Kiosk Reporting**:
    - Enables administrators to generate transaction reports at the customer or kiosk level.
    - Reports can be viewed onscreen as summaries or downloaded as detailed PDFs saved locally.

Reporting and Configuration

The application flow and various kiosk parameters are configurable through the Portal, ensuring flexibility and customization for the client's needs.

General Requirements

- **Hardware Integration**:
    - Seamless communication between the Raspberry Pi CM4, motor control board, and web application.
    - Real-time inventory tracking using ToF and optical sensors.
- **Web Portal**:
    - Customizable settings for kiosk flow and survey questions.
    - On-demand reporting with PDF generation.
- **Environmental Awareness**:
    - Promotes sustainability through educational messaging and reduced environmental impact.